Breakeven Reinvestment Rate

Understanding spot rates, forward rates, and why you can't just use r_2

1. What does "breakeven reinvestment rate" mean?

> "*What rate would I need to earn in the second year — after investing in the 1-year bond — so that I end up exactly as well off as if I had invested in the 2-year bond?*"

So we're looking for the **one-year rate, one year from now**, that makes the two strategies equivalent.

This is essentially a **1-year forward rate starting one year from now**.

2. Why can't we simply use r_2?

> **This is the biggest trap.**

The 2-year bond has coupon payments. For example, suppose:

Assumption	Value
Face value	$1,000
Coupon	5%
Maturity	2 years

The cash flows are:

Time	Cash flow
Year 1	$50
Year 2	$1,050

There is a cash flow at Year 1. Therefore, the 2-year YTM is a rate that discounts **both** cash flows:

$$P = \frac{50}{(1 + r_2)} + \frac{1{,}050}{(1 + r_2)^2}$$

So r_2 is a single rate that makes the PV of multiple cash flows equal to the bond price.

It is not necessarily the same as the true 2-year zero-coupon rate.

3. What do we actually need?

We need a rate that represents the return from **Today → Year 2** for a **single** cash flow at Year 2. That's a **zero-coupon / spot rate**. Call it z_2.

So first we need to determine z_2. That's exactly why the correct approach says:

> "*First substitute the one-year rate r_1 into the two-year bond price equation to solve for the two-year spot or zero rate z_2.*"

4. Numerical example — the setup

Suppose the market gives us:

Instrument	Detail
1-year bond	$r_1 = 4\%$
2-year coupon bond	Face value $1,000, Coupon 5%, Price $1,018.87

So the 2-year bond has cash flows: **$50 at Year 1** and **$1,050 at Year 2**.

5. First, find z_2

The bond price must equal the PV of its cash flows. We know the one-year spot rate: $z_1 = r_1 = 4\%$. Therefore:

$$1{,}018.87 = \frac{50}{1.04} + \frac{1{,}050}{(1+z_2)^2}$$

Step 1 — discount the first cash flow:

$$\frac{50}{1.04} = 48.0769$$

Step 2 — subtract from the price:

$$1{,}018.87 - 48.0769 = 970.7931$$

Step 3 — solve for z_2:

$$970.7931 = \frac{1{,}050}{(1+z_2)^2}$$

$$(1+z_2)^2 = \frac{1{,}050}{970.7931}$$

$$1 + z_2 = \left(\frac{1{,}050}{970.7931}\right)^{0.5}$$

$z_2 \approx 4.0\%$

(In this particular numerical example, the rates happen to line up conveniently.)

6. Now find the breakeven reinvestment rate

We know: $z_1 = r_1 = 4\%$, and $z_2 = 4\%$.

The relationship between spot rates and the one-year forward rate is:

$$(1+z_1)(1+f_{1,2}) = (1+z_2)^2$$

Substituting in the values:

$$(1+4\%)(1+f) = (1+4\%)^2$$

$$1 + f = 1.04$$

f = 4%

Conclusion

> *If you invest for one year at 4%, then reinvest for another year at 4%, your two-year compounded return equals the two-year*

> *zero-coupon return.*